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NPS/001/024 – Technical Specification for Low Voltage Aerial Bundled Conductor Fittings

1. Purpose

The purpose of this document is to outline the technical requirements for fittings used with low voltage Aerial Bundled Conductor overhead lines on the CE Electric UK distribution network.

This document supersedes the following documents, all copies of which should be destroyed.

Ref	Version	Title
DSS/004/019	1.0	Specification for Low Voltage Aerial Bundled Conductor Fittings
MS-O-01		Technical Specification for Aerial Bundled Conductor Fittings for Low Voltage Overhead Lines
NPS/001/024	1.0	Specification for Low Voltage Aerial Bundled Conductor Fittings

2. Scope

This specification applies to all ABC fittings and details their range of size where applicable. The majority of items are detailed in ENA TS 43-14 Issue 2: 1990.

3. Technical Requirements

3.1 General

ENA TS 43-14 Issue 2 1990 details the standard to which ABC fittings shall generally be supplied.

Appendix 2 of this network product specification provides a self certification conformance declaration that shall be completed for each item offered.

All items shall be designed to meet the performance requirements of the various sections of this specification. They must be adequately rated for their intended application and retain this rating during their normal life in an outdoor environment.

The finish shall be such that outer surfaces are free from sharp edges and burrs that could result in damage to adjacent materials or personnel.

Items comprising different component parts shall be so designed that they can be applied without disassembly.

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Materials shall be resistant to climatic stress and all insulating materials shall be UV stable. Steel components shall be hot-dip galvanised to BS EN 1461 and other metallic parts shall be of a non-corrosive type. Non-metallic materials shall be fire retardant in accordance with ENA TS 43-14, Section 4.1.3.

3.2 Specification Requirements

3.2.1 Anchor Clamps

Anchor clamps shall be designed for use with 2 and 4 core conductors in the range $35\text{mm}^2 - 120\text{mm}^2$. They shall be provided in two classes: Heavy duty for conductors operating at full line tension and light duty for conductors operating at a reduced tension for services attached to buildings. Full tension 4-core anchor clamps shall be range taking $35\text{mm}^2 - 120\text{mm}^2$ ABC.

Anchor clamps shall be designed for attachment to hook and eye bolts as detailed in ENA TS 43-13 Section 5.

Details of the range of fittings required are listed in Appendix 1.

3.2.2 Suspension Clamps

The clamp shall be designed to allow the ABC to be erected without the use of a traditional conductor running blocks and to be clamped in place on completion. The clamp shall be suitable for angles of deviation up to 60° when using 2 or 4-core $35\text{mm}^2 - 120\text{mm}^2$ ABC. The clamp shall incorporate a means of securing the conductors in place. Suspension clamps shall be tested in accordance with ENA TS 43-14 Section 7.3.

Details of the range of fittings required are listed in Appendix 1.

3.2.3 Branch and Service Connections

The connector shall be of a type, which has been designed for application to the main conductors without the removal of the insulation (insulation piercing), and simultaneously pierce the insulation of the ABC main and tap conductors. The connector shall be waterproof in accordance with category IP68 and satisfy the degree of protection requirements as specified in BS EN 60529. The design must incorporate an end cap that shall be suitable for location on either side of the connector. The connectors shall be suitable for installation without any special tooling and supplied with bolts designed to shear at the required torque. The connectors shall be range taking where possible and suitable for ABC connections to ABC, ABC to service cables or bare Copper and Plain Aluminium conductors.

Details of the range of fittings required are listed in Appendix 1.

3.2.4 Mains Earth Fitting

Shall be manufactured to the same standard as branch and service IPC connectors detailed in 3.2.3 but incorporate the facility for testing, application of earthing lead or generator connection. They shall be range taking and suitable for use on 35mm^2 to 120mm^2 ABC.

3.2.5 Cable End Cap

End caps shall be insulated and provide protection against the ingress of moisture and also ensure that the system is fully insulated. They shall maintain the degree of protection offered by IP68 in accordance with BS EN 60529. The caps shall be permanent cold applied, range taking and suitable for installation on $35\text{mm}^2 - 120\text{mm}^2$ ABC.

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3.2.6 Conductor Joint Insulation Sleeve

Shall be suitable for application over compression joints associated with 35mm² – 120mm² ABC. The sleeve shall be suitable for application without the use of heat or special tooling. It shall protect against the ingress of moisture and restore system insulation properties.

3.2.7 Outdoor Aerial Service Cable Distribution Box

The enclosure shall provide a degree of protection in accordance with category IP43 as specified in BS EN 60529 with suitable drainage provided to clear any ingress of moisture. The lid shall be secured to the box of the main body when open and the box shall be suitable for installation on wood or steel poles. The main supply connection to the box shall accept 4 x 35mm², 50mm² or 70mm² ABC. The box shall have provision for approximately nine single-phase copper or plain aluminium phase and neutral conductors with a maximum cross sectional area of 35mm². Terminal connections shall be suitable for installation without any special tooling.

The enclosure shall provide the facility for the connection of a 32mm² PVC copper PME earth lead to the neutral block.

3.2.8 Proximity Shrouding

Wall mounted proximity shrouding shall be suitable for use on 4-core 35mm² – 120mm² ABC and supplied in 2.5 metre lengths. The material shall be non-rigid, extruded black tube with a wall thickness of 1.5mm.

3.2.9 Cleats

Wall Mounted

Wall mounted cleats shall provide 100mm of stand off and be capable of carrying 35mm² – 120mm² ABC. They shall be designed to be secured to the fabric of the building and include an integral means of securing the cable.

Pole Mounted

Pole cleats shall be suitable for mounting on steel or wood poles. They shall be range taking and accept 35mm² – 120mm² ABC. For steel poles steel bandit tape and buckles shall be supplied with each cleat.

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4. References

4.1 External Documentation

Reference	Title
ENA TS 43-14 Issue 2 1990	Conductor Fittings and Associated Apparatus for use on LV Aerial Bundled Conductor
BS EN ISO 1461: 1999	Hot dipped galvanised coating on fabricated iron or steel articles. Specification and test methods.
BS EN 60529: 1992	Degrees of Protection Provided by Enclosures. (IP rating)

4.2 Internal documentation

Reference	Title
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5. Definitions

Term	Definition
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6. Authority for issue

6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
S Salkeld	Standards Engineer	S Salkeld	31/08/07

6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
G Bolden	Purchasing Manager	G Bolden	04/09/07

6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
R McMahon	Standards Engineer	R McMahon	31/08/07

6.4 DBD Assurance

I sign to confirm that this document has been assured for issue on to the DBD system

		Sign	Date
S Johnson	DBD Administrator	S Johnson	21/09/07

6.5 Approval

Approval is given for the content of this document

		Sign	Date
C Holdsworth	Standards and Assessment Manager	C Holdsworth	31/08/07

6.6 Authorisation

Authorisation is granted for publication of this document

		Sign	Date
C Morgan	Head of Procurement	C Morgan	05/09/07

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Appendix 1 – Schedule of Components

Commodity Code	Description	Clause
242359	Anchor Clamp, Full Tension 50mm ² – 120mm ² 4-core	3.2.1
220772	Anchor Clamp, Full Tension 35mm ² 4-core	3.2.1
221085	Anchor Clamp, Full Tension 35mm ² 2-core	3.2.1
221069	Anchor Clamp, Reduced Tension 1.4KN 35mm ² 4-core	3.2.1
221077	Anchor Clamp, Reduced Tension 1.4KN 35mm ² 2-core	3.2.1
242217	Combined Running Block/Suspension Clamp 50mm ² – 120mm ² 2 and 4-core	3.2.2
242132	Insulation Piercing Connector, Single Bolt 50mm ² -15mm ² main to 6mm ² -35 mm ² tap	3.2.3
222307	Insulation Piercing Connector, Single Bolt 25mm ² -95 mm ² main to 6mm ² -35mm ² tap	3.2.3
222323	Insulation Piercing Connector, Double Bolt 50mm ² -150mm ² main to 6mm ² -35mm ² tap	3.2.3
241996	Insulation Piercing Connector, Double Bolt 50mm ² -185mm ² main to 50mm ² -150mm ² tap	3.2.3
242024	Insulation Piercing Connector, Double Bolt for 35mm ² – 120mm ² ABC Main to O/W Conductors (32mm ² – 100mm ² Cu. or 50mm ² – 100mm ² Plain Al.)	3.2.3
222364	Insulation Piercing Connector, Earthing Fitting for 35mm ² to 120 mm ² ABC	3.2.4
265524	Cable End Cap for 35mm ² – 120mm ² ABC	3.2.5
262210	Joint Insulation Sleeve	3.2.6
262028	ABC Service Distribution Box	3.2.7
265774	Proximity Shrouding, 2.5 metre Length	3.2.8
262051	Wall Mount Cleat	3.2.9
242560	Pole Cleat	3.2.9

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Appendix 2

SELF CERTIFICATION CONFORMANCE DECLARATION

Aerial Bundled Conductor Fittings covered by ENATS 43-14 shall comply with the latest issues of the relevant international and British Standards.

ENATS 43-14 is intended to amplify and/or clarify the requirements of those Standards.

This check sheet identifies the clauses in ENATS 43-14 and the clauses of the aforementioned Standards relevant to ABC Fittings for use on current United Kingdom designs of ABC low voltage overhead lines.

The manufacturer shall declare conformance or otherwise, clause by clause, using the following levels of conformance declaration codes.

Conformance declaration codes

N/A = Clause is not applicable/ appropriate to the product

Cs1 = The product conforms fully with the requirements of this clause

Cs2 = The product conforms partially with the requirements of this clause

Cs3 = The product does not conform to the requirements of this clause

Cs4 = The product does not currently conform to the requirements of this clause, but the manufacturer proposes to modify and test the product in order to conform.

Instructions for completion

- When Cs1 code is entered no remark is necessary
- When any other code is entered the reason for non-conformance shall be entered
- Prefix each remark with the relevant 'BS EN' or 'ENATS' as appropriate

Manufacturer:

Product Reference:

Name:

Signature:

Date:

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Appendix 2 (continued)

Aerial Bundled Conductor Fittings

ENA TS 43-14 Issue 2, 1990			
Clause/Sub-clause	Requirement	Conformance Code	Remarks
Section 4.1	General Requirements for Materials		
Section 4.2	General Requirements for Tests		
Section 4.4	Identification and Markings		
Section 6.1.3	Test for Wall Mounted Cleats		
Section 7.3	Tests for Anchor Clamps		
Section 8.3	Tests for Suspension Clamps		
Section 9.4	Tests for Insulation Piercing Connectors (waterproof to IP68 as 9.5.5b)		
Section 11.4	Labels for Service Distribution Boxes		
Section 11.5	Tests for Service Distribution Boxes		
Section 12.2.3	Tests for Proximity Shrouding and Cable End Caps		
Section 12.2.3	Tests Applicable to Conductor Joint Insulation Sleeve		